

CARLOS CASTILLO DOMÍNGUEZ

Software Engineer

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PROFESSIONAL PROFILE

Computer Science Engineer with hands-on experience in software development, microservices architectures, and SaaS systems. Contributed to scalable projects using full-stack technologies (JavaScript, TypeScript, NestJS, React), including SQL/NoSQL data modeling, REST API integration, functional UI development, and cloud deployment. Results-oriented professional with strong analytical skills, adaptability to new environments, and experience collaborating in Agile teams using Kanban and Scrum methodologies. Motivated to continuously learn, take on new challenges, and strengthen professional growth through real-world, high-impact projects.

EXPERIENCE

Web Developer – Capstone Project

Aug 2024 – Jul 2025

University of Santiago of Chile, Chile

- Led the development of a SaaS-based CMS platform for managing meeting websites, using Kanban (Trello) to prioritize client requirements and project tasks.
- Built an architecture composed of 10 microfrontends (React) integrated with a REST API of 6 microservices (Node.js/NestJS) using a NoSQL model (MongoDB), ensuring long-term CMS scalability.
- Managed source code with Git/GitHub (CI/CD) and deployment in a virtualized Docker environment, achieving 80% test coverage with SonarQube.
- Reduced implementation time by 90% for three new meeting management systems.

IT Services Intern

Jan 2025 – Feb 2025

Comercializadora de Trigo S.A., Chile

- Performed corrective and evolutionary maintenance on the corporate website using Django under the MVC pattern (PostgreSQL, Python), resolving critical issues and ensuring operational continuity.
- Designed and developed a centralized corporate video repository, improving public visibility and accessibility of previously scattered audiovisual content.
- Redesigned the Whistleblower Channel module, improving the visualization of critical data by 30%.
- Resolved five issues within the SQL data model (MySQL) of the logistics system, maintaining uninterrupted operational continuity for end users.

User Experience Teaching Assistant – UX

Aug 2024 – Dec 2024

University of Santiago of Chile, Chile

- Centralized academic resources across Google Sites, Drive, and GitHub, reducing resource search time to under five minutes.
- Evaluated two educational software platforms and created user manuals, reducing technical support requests by 30%.
- Collaborated on the design of the final course project, mentoring a team of five students under the Scrum methodology.
- Delivered structured feedback throughout five project iterations, increasing the quality of final deliverables by 30% compared to previous cohorts.

Software Engineer – Computer Engineering Project

Aug 2023 – Dec 2023

University of Santiago of Chile, Chile

- Participated in the development of a document management application under Scrum (Jira), leveraging advanced Python libraries to process PDF documents stored locally and in Google Drive.
- Contributed to the implementation of an advanced search engine using NLTK and PyPDF, reducing document retrieval time for unstructured repositories.
- Assisted in the implementation of a document classification system using Scikit-Learn, automating document organization based on date, language, and topic criteria.
- Supported the implementation of an AI-powered chatbot using OpenAI for document interaction, reducing response time for frequently asked questions.

Frontend Developer – Software Engineering Workshop

Mar 2023 – Jul 2023

University of Santiago of Chile, Chile

- Contributed to the development of a web platform designed to automate course management and interactive number-line exercise simulations.
- Developed the user interface using Vue.js while integrating a REST API built with Spring Boot/Java over a PostgreSQL relational model, improving user interaction through functional views.
- Achieved 80% test coverage with SonarQube, including unit and integration testing.
- Reduced the operational workload of teaching staff by 30% in exercise generation tasks.

EDUCATION

B.Sc. in Computer Engineering (Graduated)

Mar 2018 – Jul 2025

University of Santiago of Chile, Chile

Coursework: Programming Paradigms, Software Engineering Fundamentals, Database Workshop, Software Engineering Workshop, User Experience (UX).

TECHNICAL SKILLS

- **Programming Languages:** TypeScript, Python, Java.
- **Frontend:** HTML5, CSS3, JavaScript, React, Vue.js, Bootstrap, TailwindCSS.
- **Backend:** Node.js, NestJS, Django, Spring Boot.
- **Databases:** MySQL, PostgreSQL, MongoDB.
- **DevOps & Tools:** Git, GitHub, Linux, Docker, AWS, Vercel, CI/CD, Figma, Postman.
- **Agile Methodologies:** Kanban (Trello), Scrum (Jira).
- **Languages:** Spanish (Native), English (Intermediate).